ILLINOIS INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

Database Organization
Course CS-425 02

Homework No.6
Optional Homework
SQL Advanced window functions & OLAP queries
SalingDB

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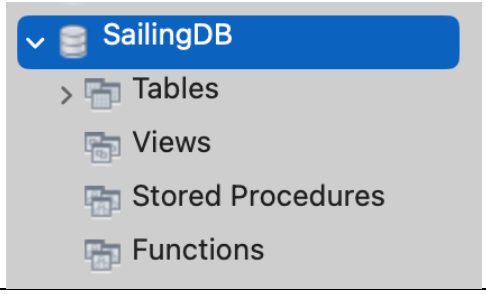
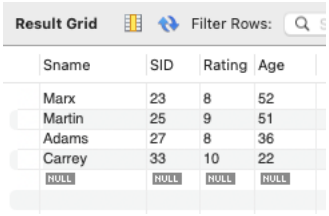
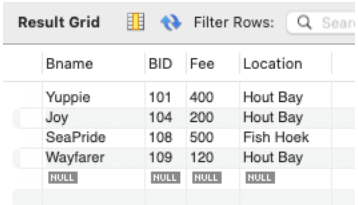
Consider a Sailing Club Database with Schema:
Boats-schema = (BID, Bname, fee, location)
Sailors-schema = (SID, Sname, rating, age)
Reserves-schema = (SID, BID, day, deposit)
Captains-schema = (SID, Sname, rating, age)

DATABASE CREATION

SQL CODE	CAPTURE
CREATE DATABASE **IF NOT EXISTS** SailingDB; **USE** SailingDB;	

TABLES AND VALUES INSERTION

SQL CODE	CAPTURE
CREATE TABLE sailors (Sname **varchar**(20), SID **int NOT NULL**, Rating **int**, Age **int**, **PRIMARY KEY** (SID)); INSERT **INTO** sailors(Sname,SID,Rating,Age) VALUES('Marx',23,8,52), ('Martin',25,9,51), ('Adams',27,8,36), ('Carrey',33,10,22);	 Sname SID Rating Age Marx 23 8 52 Martin 25 9 51 Adams 27 8 36 Carrey 33 10 22 NULL NULL NULL NULL

CREATE TABLE boats(Bname **varchar**(20), BID **int NOT NULL**, Fee **int**, Location **text**(20), **PRIMARY KEY**(BID)); INSERT **INTO** boats(Bname,BID,Fee,Location) VALUES ('Wayfarer',109,120,'Hout Bay'), ('SeaPride',108,500,'Fish Hoek'), ('Yuppie',101,400,'Hout Bay'), ('Joy',104,200,'Hout Bay');	 Bname BID Fee Location Yuppie 101 400 Hout Bay Joy 104 200 Hout Bay SeaPride 108 500 Fish Hoek Wayfarer 109 120 Hout Bay NULL NULL NULL NULL

SQL CODE	CAPTURE
CREATE TABLE reserves(SID **INT**, BID **INT**, Day DATE, Deposit **DECIMAL**(10, 2), **PRIMARY KEY** (SID, BID), **FOREIGN KEY** (SID) **REFERENCES** sailors(SID), **FOREIGN KEY** (BID) **REFERENCES** boats(BID)); INSERT **INTO** reserves (SID, BID, Day, Deposit) VALUES (23, 109, '2014-08-01', 120.00), (23, 108, '2014-08-08', 120.00), (25, 101, '2014-08-08', 0.00), (27, 101, '2014-08-09', 100.00), (27, 109, '2014-08-15', 120.00), (33, 109, '2014-09-04', 0.00), (33, 104, '2014-09-11', 0.00);	Result Grid Filter Rows: Sear SID BID Day Deposit 23 108 2014-08-08 120.00 23 109 2014-08-01 120.00 25 101 2014-08-08 0.00 27 101 2014-08-09 100.00 27 109 2014-08-15 120.00 33 104 2014-09-11 0.00 33 109 2014-09-04 0.00 NULL NULL NULL NULL

CREATE TABLE captains (Sname **varchar**(20), SID **int NOT NULL**, Rating **int**, Age **int**, **PRIMARY KEY** (SID));	Result Grid Filter Rows: Sname SID Rating Age NULL NULL NULL NULL

DATABASE CONTEXT. QUESTIONS

1. **Get everything in the Sailors table.**

SQL QUERY	CAPTURE
SELECT * **FROM** sailors;	Result Grid Filter Rows: Search Sname SID Rating Age Marx 23 8 52 Martin 25 9 51 Adams 27 8 36 Carrey 33 10 22 NULL NULL NULL NULL

2. **Get sailor ID, rank & age of all sailors, ordered from highest to lowest rank. Rank is 10 times rating.**

SQL QUERY	CAPTURE
SELECT SID, Rating * 10 **AS** 'Rank', Age **FROM** sailors **ORDER BY** Rating **DESC**;	Result Grid Filte SID Rank Age 33 100 22 25 90 51 23 80 52 27 80 36

3. **Get alphabetical list of sailors with rating less than 10.**

SQL QUERY	CAPTURE
SELECT Sname **FROM** sailors **WHERE** Rating < 10 **ORDER BY** Sname;	Result Grid Filter Rows: Search Sname Adams Martin Marx

4. **Find how much deposit money there is in total and how many tuples are in the reserves table.**

SQL QUERY	CAPTURE
SELECT SUM(Deposit) **AS** TotalDeposit, COUNT(*) **AS** TupleCountReserves **FROM** reserves;	Result Grid Filter Rows: TotalDeposit TupleCountReserv... 460.00 7

5. Get all info on boats in Fishhoek but I'm not sure how you spell Fishoek.

SQL QUERY	CAPTURE
SELECT * **FROM** boats **WHERE** Location **LIKE** '%Fish Hoek%'	Result Grid Filter Rows: Search Bname BID Fee Location SeaPride 108 500 Fish Hoek NULL NULL NULL NULL

6. In what locations are boats kept?

SQL QUERY	CAPTURE
SELECT DISTINCT Location **FROM** boats;	Result Grid Filter Rows: Search Location Hout Bay Fish Hoek

7. Get the names of all Boats that have a fee value recorded in the database.

SQL QUERY	CAPTURE
SELECT Bname **FROM** boats **WHERE** Fee **IS NOT NULL**;	Result Grid Filter Rows: Search Bname Yuppie Joy SeaPride Wayfarer

8. Get ID of all boats that have not been reserved.

SQL QUERY	CAPTURE
#NOTE. *THIS QUERY IS SHOWING THE IDs WITH A DEPOSIT EQUAL TO 0 IN THE RESERVATION TABLE.* **SELECT** BID **FROM** reserves **WHERE** Deposit =0; ***#NOTE.*** *THIS QUERY IS RETURNING A BLANK VALUE BECAUSE IT IS TAKING ALL IDs FROM THE RESERVATION VALUE AS RESERVED.* SELECT boats.BID FROM boats LEFT JOIN reserves ON boats.BID = reserves.BID WHERE reserves.BID IS NULL;	Result Grid Filter Rows: Se BID 101 104 109 Result Grid Filter R BID

9. Get all reservation info, including all details on the boats being reserved.

SQL QUERY	CAPTURE
SELECT * **FROM** reserves **JOIN** boats **ON** reserves.BID = boats.BID;	Result Grid Filter Rows: Search Export:

SID	BID	Day	Deposit	Bname	BID	Fee	Location
25	101	2014-08-08	0.00	Yuppie	101	400	Hout Bay
27	101	2014-08-09	100.00	Yuppie	101	400	Hout Bay
33	104	2014-09-11	0.00	Joy	104	200	Hout Bay
23	108	2014-08-08	120.00	SeaPride	108	500	Fish Hoek
23	109	2014-08-01	120.00	Wayfarer	109	120	Hout Bay
27	109	2014-08-15	120.00	Wayfarer	109	120	Hout Bay
33	109	2014-09-04	0.00	Wayfarer	109	120	Hout Bay

10. For all reservations, get the name of the sailor, along with the day and name of the boat booked.

SQL QUERY	CAPTURE
SELECT Sname, Day, Bname **FROM** sailors **JOIN** reserves **ON** sailors.SID = reserves.SID **JOIN** boats **ON** reserves.BID = boats.BID;	Result Grid Filter Rows: Search

Sname	Day	Bname
Marx	2014-08-08	SeaPride
Marx	2014-08-01	Wayfarer
Martin	2014-08-08	Yuppie
Adams	2017-08-09	Yuppie
Adams	2014-08-15	Wayfarer
Carrey	2014-09-11	Joy
Carrey	2014-09-04	Wayfarer

11. Get the average deposit paid for each boat.

SQL QUERY	CAPTURE
SELECT BID, **ROUND(AVG**(Deposit),2) AS 'Avg_Deposit' **FROM** reserves **GROUP BY** BID;	Result Grid Filte

BID	Avg_Deposit
101	50.00
104	0.00
108	120.00
109	80.00

12. Get the average deposit paid for each boat that has been booked by more than one person.

SQL QUERY	CAPTURE
SELECT BID, ROUND(AVG(Deposit),2) AS 'Avg_Deposit' FROM reserves GROUP BY BID HAVING COUNT(DISTINCT SID) > 1;	Result Grid F

BID	Avg_Deposit
101	50.00
109	80.00

13. Get the average firm deposit paid for each boat that has been booked by more than one person, in increasing order of amount. A firm deposit is one which exceeds R10.

SQL QUERY	CAPTURE
SELECT BID, ROUND(AVG(Deposit),2) AS 'Avg_Deposit' **FROM reserves** **WHERE Deposit > 10** **GROUP BY BID** **HAVING COUNT(DISTINCT SID) > 1** **ORDER BY 'Avg_Deposit';**	Result Grid Filter Rows: BID Avg_Deposit 109 120.00

14. Get name & rating of sailors with rating exceeding 7 who made any reservation with 0 deposit.

SQL QUERY	CAPTURE
SELECT Sname, Rating **FROM** sailors **WHERE** Rating > 7 **AND** SID **IN** (**SELECT** SID **FROM** reserves **WHERE** Deposit = 0);	Result Grid Filter R Sname Rating Martin 9 Carrey 10

15. Get names of boats located in a place other than Hout Bay or Fish Hoek.

SQL QUERY	CAPTURE
SELECT Bname **FROM** boats **WHERE** Location **NOT IN** ('Hout Bay', 'Fish Hoek');	Result Grid Filter Rows: Search Export: Bname

16. Get names of boats having a fee larger than any boat located in Hout Bay.

SQL QUERY	CAPTURE
SELECT DISTINCT Bname **FROM** boats **WHERE** Fee > (**SELECT** MAX(Fee) **FROM** boats **WHERE** Location = 'Hout Bay');	Result Grid Filter Bname SeaPride

17. Get names that are in both the sailors and the captains relations.

SQL QUERY	CAPTURE
SELECT sailors.Sname **FROM sailors** **INNER JOIN captains ON sailors.Sname = captains.Sname;**	Result Grid Filter Rows: Search Export: Sname

18. Get names of boats that have exactly 1 reservation.

SQL QUERY	CAPTURE
SELECT Bname **FROM** boats **WHERE** BID **IN** (**SELECT** BID **FROM** reserves **GROUP BY** BID **HAVING** COUNT(*) = 1);	Result Grid F Bname Joy SeaPride

19. Get sailor ID and total deposit paid for sailors who have booked more than 1 boat.

SQL QUERY

#NOTE. *THIS QUERY REPRESENTS ALL RESERVATION INCLUDING THE 0 DEPOSIT.*

```
SELECT SID, SUM(Deposit) AS TotalDeposit
FROM reserves
GROUP BY SID HAVING COUNT(BID) > 1;
```

#NOTE. *THIS QUERY REPRESENTS ALL RESERVATION EXCLUSING THE 0 DEPOSIT.*

```
SELECT SID, SUM(Deposit) AS TotalDeposit
FROM reserves
WHERE Deposit > 0
GROUP BY SID
HAVING COUNT(DISTINCT BID) > 1;
```

CAPTURE

Result Grid Filter R

SID	TotalDeposit
23	240.00
27	220.00
33	0.00

Result Grid Filter Rows:

SID	TotalDeposit
23	240.00
27	220.00

20. Get all reservation info including details of the boat booked.

SQL QUERY

```
SELECT *
FROM reserves
INNER JOIN boats ON reserves.BID = boats.BID;
```

CAPTURE

Result Grid Filter Rows: Search Export:

SID	BID	Day	Deposit	Bname	BID	Fee	Location
25	101	2014-08-08	0.00	Yuppie	101	400	Hout Bay
27	101	2014-08-09	100.00	Yuppie	101	400	Hout Bay
33	104	2014-09-11	0.00	Joy	104	200	Hout Bay
23	108	2014-08-08	120.00	SeaPride	108	500	Fish Hoek
23	109	2014-08-01	120.00	Wayfarer	109	120	Hout Bay
27	109	2014-08-15	120.00	Wayfarer	109	120	Hout Bay
33	109	2014-09-04	0.00	Wayfarer	109	120	Hout Bay

21. Get all information on every boat. If a boat has reservations, including all its reservations info.

SQL QUERY

```
SELECT * FROM boats
LEFT JOIN reserves ON boats.BID = reserves.BID;
```

CAPTURE

Result Grid Filter Rows: Search Export:

Bname	BID	Fee	Location	SID	BID	Day	Deposit
Yuppie	101	400	Hout Bay	25	101	2014-08-08	0.00
Yuppie	101	400	Hout Bay	27	101	2014-08-09	100.00
Joy	104	200	Hout Bay	33	104	2014-09-11	0.00
SeaPride	108	500	Fish Hoek	23	108	2014-08-08	120.00
Wayfarer	109	120	Hout Bay	23	109	2014-08-01	120.00
Wayfarer	109	120	Hout Bay	27	109	2014-08-15	120.00
Wayfarer	109	120	Hout Bay	33	109	2014-09-04	0.00

22. Create a new tuple for the boat named "Nino" which has fee R150, BID 110, and is in Fish Hoek.

SQL QUERY

```
INSERT INTO boats (Bname, BID, Fee, Location)
VALUES ('Nino', 110, 150, 'Fish Hoek');
SELECT *
FROM SailingDB.boats;
```

CAPTURE

Result Grid Filter Rows: Search

Bname	BID	Fee	Location
Yuppie	101	400	Hout Bay
Joy	104	200	Hout Bay
SeaPride	108	500	Fish Hoek
Wayfarer	109	120	Hout Bay
Nino	110	150	Fish Hoek
NULL	NULL	NULL	NULL

23. Remove all bookings from Reserves where there is no deposit.

SQL QUERY	CAPTURE
SET SQL_SAFE_UPDATES = 0; DELETE FROM reserves WHERE Deposit = 0; SET SQL_SAFE_UPDATES = 1; SELECT * FROM SailingDB.reserves;	Capture Result Grid Filter Rows: Search Edi SID BID Day Deposit 23 108 2014-08-08 120.00 23 109 2014-08-01 120.00 27 101 2014-08-09 100.00 27 109 2014-08-15 120.00 NULL NULL NULL NULL

24. Increase the fee of every boat by 12%.

SQL QUERY	CAPTURE
SET SQL_SAFE_UPDATES = 0; UPDATE boats SET Fee = Fee * 1.12; SET SQL_SAFE_UPDATES = 1; SELECT * FROM SailingDB.boats;	Result Grid Filter Rows: Search E Bname BID Fee Location Yuppie 101 448 Hout Bay Joy 104 224 Hout Bay SeaPride 108 560 Fish Hoek Wayfarer 109 134 Hout Bay Nino 110 168 Fish Hoek NULL NULL NULL NULL

25. Make a view called Bookings which hides the Deposit value i.e. only has the other 3 attributes.

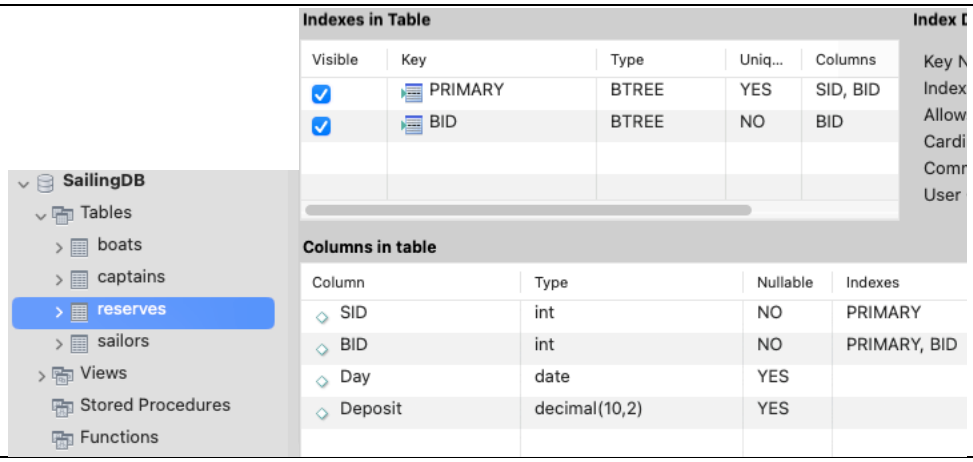
SQL QUERY	CAPTURE
CREATE VIEW Bookings **AS SELECT** SID, BID, Day **FROM** reserves; SELECT * FROM SailingDB.bookings;	SailingDB Tables boats captains reserves sailors Views bookings Stored Procedures Functions Result Grid Filter Rov SID BID Day 23 108 2014-08-08 23 109 2014-08-01 27 101 2014-08-09 27 109 2014-08-15

26. Create a table called Reserves with 3 integer attributes BID, SID & deposit, and a date attribute Day. Allow only deposit to be omitted, and ensure SID and BID values exist in the database. When someone books a boat, it is for the whole day.

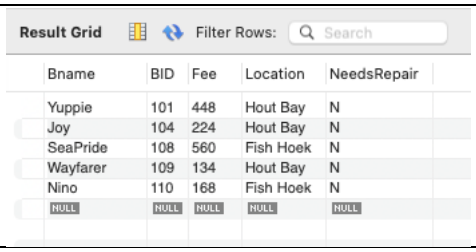
SQL QUERY

```
CREATE TABLE reserves (
   BID INT NOT NULL,
   SID INT NOT NULL,
   Day DATE NOT NULL,
   Deposit INT,
   PRIMARY KEY (BID, SID),
   FOREIGN KEY (BID) REFERENCES boats(BID),
   FOREIGN KEY (SID) REFERENCES sailors(SID)
);
```

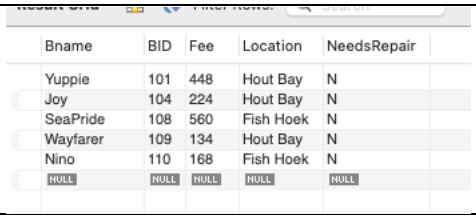
CAPTURES



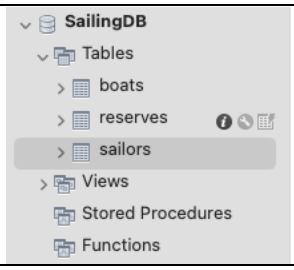
27. Add a new attribute NEEDSREPAIR to the Boats table. It is usually "N".

SQL QUERY	CAPTURE
ALTER TABLE boats ADD NeedsRepair CHAR(1) DEFAULT 'N'; SELECT * FROM SailingDB.boats;	

28. We should not be ageist. Remove the Age attribute.

SQL QUERY	CAPTURE
ALTER TABLE sailors DROP COLUMN Age; **SELECT * FROM SailingDB.boats;**	

29. Remove the Captains relation altogether so that nobody can try insert or use Captains in future.

SQL QUERY	CAPTURE
DROP TABLE captains;	

RELATION ALGEBRA

#2 **Get sailor ID, rank & age of all sailors, ordered from highest to lowest rank. Rank is 10 times rating.**

SQL QUERY	RELATION ALGEBRA
SELECT SID, Rating * 10 **AS** 'Rank', Age **FROM** sailors **ORDER BY** Rating **DESC**;	$\pi_{SID,\ (Rating\ X\ 10)\ 'Rank',\ Age}\ \sigma\ ((sailors))$

#8 Get ID of all boats that have not been reserved.

SQL QUERY	RELATION ALGEBRA
SELECT BID **FROM** reserves **WHERE** Deposit <=0; SELECT boats.BID FROM boats LEFT JOIN reserves ON boats.BID = reserves.BID WHERE reserves.BID IS NULL;	$\pi_{BID}(\sigma_{Deposit <=0}((reserves))$ $\pi_{BID}(\sigma_{reserves.BID\ IS\ NULL}(boats \ltimes_{boats.BID=reserves.BID} reserves))$

#9 Get all reservation info, including all details on the boats being reserved.

SQL QUERY	RELATION ALGEBRA
SELECT * **FROM** reserves **JOIN** boats **ON** reserves.BID = boats.BID;	$reserves \bowtie_{reserves.BID = boats.BID} (boats)$

#10 For all reservations, get the name of the sailor, along with the day and name of the boat booked.

SQL QUERY	RELATION ALGEBRA
SELECT Sname, Day, Bname **FROM** sailors **JOIN** reserves **ON** sailors.SID = reserves.SID **JOIN** boats **ON** reserves.BID = boats.BID;	$\pi_{Sname,\ Day,\ Bname}((sailors \bowtie_{sailors.ID = reserves.ID} (reserves)) \bowtie_{reserves.BID = boats.BID} (boats))$

#17 Get names that are in both the sailors and the captains relations.

SQL QUERY	RELATION ALGEBRA
SELECT sailors.Sname **FROM sailors** **INNER JOIN captains ON sailors.Sname = captains.Sname;**	$\pi_{sailors.Sname}$ $(salilors \bowtie_{sailors.Sname = captains.Sname} (captains)$

#20

SQL QUERY	RELATION ALGEBRA
SELECT * **FROM reserves** **INNER JOIN boats ON reserves.BID = boats.BID;**	$reserves \bowtie_{reserves.BID = boats.BID} (boats)$